

△CS Layer 5 — Cross-Discipline Generative Layer

Triangulated Coherent Systems (△CS) Governance Documentation

Document ID:	△CS-GOV-L5-CDGL	Version:	1.0
Status:	Active	Effective Date:	July 2026
Classification:	Governance Framework — Restricted	Layer Designation:	Cross-Discipline Generative Layer (CDGL)
Supersedes:	None (inaugural issuance)	Next Review:	July 2027

Document Purpose

This document governs the conditions, protocols, and stewardship mechanisms under which genuine cross-discipline generation occurs — the production of outputs, insights, frameworks, and decisions that no single discipline could produce alone. Layer 5 is the apex of the cross-discipline stack, synthesizing coherence, synchronization, and alignment into system-level generative output ready for governance consideration.

Table of Contents

1.	Layer Identity
2.	Purpose and Scope
3.	Core Functions
4.	Structural Components
5.	Operating Principles
6.	Generative Authorization Protocol
7.	Integration Methods
8.	Interaction Rules
9.	Cadence and Review
10.	Roles and Accountabilities
11.	Escalation Pathways
12.	Amendment Conditions

1. Layer Identity

Attribute	Definition
Layer Designation	Cross-Discipline Generative Layer (CDGL)
Altitude	Layer 5 — Apex of the Cross-Discipline Stack (highest cross-discipline altitude)
Function	Governs the conditions, protocols, and stewardship of cross-discipline generative activity — producing outputs that transcend any single discipline's capacity.
Layer Above	Governance-Level Evaluation Layer (Layer 6) — receives all GOC-cleared generative outputs
Layer Below	Cross-Discipline Alignment Layer (Layer 4) — delivers alignment verification upstream
Stack Position	Receives the convergent outputs of all cross-discipline layers (Layers 2-4) and

Attribute	Definition
	channels them into governance-ready generative production.

Stack Context Note

The CDGL sits at the productive apex of the cross-discipline stack. It does not operate independently of lower layers: it is the culminating structure through which coherence (Layer 2), synchronization (Layer 3), and alignment (Layer 4) are converted into formal generative output. No generative initiative may proceed without confirmed prerequisites from each of these layers.

2. Purpose and Scope

2.1 Purpose

The Cross-Discipline Generative Layer (CDGL) represents the productive summit of the cross-discipline stack — the layer at which aligned, coherent, and synchronized disciplines actively generate together. Its purpose is to authorize, structure, steward, and certify all cross-discipline generative activity within the Δ CS framework, ensuring that every generative output submitted to governance is of verified integrative origin and demonstrated governance utility.

2.2 Definition of Generative Output

For the purposes of this document, **generative output** is defined as any deliverable, decision-input, innovation, or framework that satisfies both of the following conditions:

1. It requires the genuine integration of two or more disciplines in its production; and
2. It could not have been produced at full quality within any single discipline operating alone.

Outputs satisfying only condition (a) but not condition (b) are classified as *discipline-adjacent outputs* and are not subject to CDGL governance. Classification is determined by the Generative Director at the point of initiative review.

2.3 Scope

This document and the protocols herein apply to:

- All cross-discipline generative initiatives conducted within the Δ CS framework;
- All joint deliverables produced through the collaboration of two or more disciplines;
- All integration projects designated as cross-discipline in nature by Layers 2, 3, or 4;
- All emergent outputs arising from authorized generative processes; and
- All outputs submitted to governance bearing a cross-discipline designation.

3. Core Functions

The CDGL performs four formally designated core functions. Each function is governed by its own standards, protocols, and instruments as specified below.

3.1 Generative Initiative Authorization

The CDGL authorizes and registers all cross-discipline generative initiatives before substantive work begins. No initiative may proceed to the generative phase without a valid Generative Authorization Notice (GAN) issued by the CDGL.

Authorization requires evaluation against the **Generative Readiness Standard (GRS)**, which mandates:

- **Coherence confirmed** — Layer 2 (Coherence Layer) readiness confirmation received;
- **Synchronization cleared** — Layer 3 (Synchronization Layer) clearance received; and

- **Alignment verified** — Layer 4 (Alignment Layer) alignment verification received.

Upon GRS confirmation, the CDGL issues a **Generative Authorization Notice (GAN)** to all initiating disciplines and registers the initiative in the Generative Initiative Register (GIR).

3.2 Integration Architecture

The CDGL designs and maintains the **Integration Architecture** for each authorized initiative. The Integration Architecture is the structural plan specifying how contributing disciplines will combine their respective inputs into joint generative outputs.

Each Integration Architecture Document (IAD) shall specify, at minimum:

- The contributing disciplines and their designated roles;
- The selected integration method (see Section 7);
- Quality gates applicable at defined generative milestones;
- The required submission format for completed outputs; and
- The governance routing path upon GOS clearance.

IADs are produced in consultation with contributing disciplines and remain under CDGL authority for the life of the initiative. Amendments to an active IAD require Generative Director sign-off.

3.3 Generative Output Stewardship

The CDGL receives all completed joint outputs and evaluates them against the **Generative Output Standard (GOS)** before submission to governance.

Submission to any governance body without a valid Generative Output Certificate (GOC) is not permitted.

The GOS requires that all four of the following criteria are satisfied:

- **Cross-discipline authorship verifiable** — The output demonstrably reflects the contributions of two or more disciplines;
- **Disciplinary contributions attributed** — Each discipline's contribution is explicitly documented within or appended to the output;

- **Integration logic documented** — The method and rationale by which disciplinary contributions were integrated is recorded; and
- **Governance utility demonstrated** — The output addresses a defined governance need or decision-support requirement.

Upon satisfying all GOS criteria, the CDGL issues a **Generative Output Certificate (GOC)**, which is required for governance consideration at Layer 6.

3.4 Emergence Monitoring

The CDGL monitors all active generative processes for emergent insights, frameworks, or structural patterns not anticipated in the original Integration Architecture. Emergence is an expected feature of high-integration cross-discipline work and is treated as an institutional asset rather than an anomaly.

Emergence monitoring protocols require:

- Capture of all emergent outputs in the **Emergence Register (ER)** with governance routing notes;
- Formal distinction, in all CDGL reporting, between *planned generative output* and *emergent generative output*; and
- Immediate escalation to Layer 6 via an Emergence Alert where emergent outputs carry direct governance implications.

4. Structural Components

The CDGL operates through six formally defined structural components. Each component is a living instrument maintained by the CDGL and is subject to amendment under the conditions specified in Section 12.

Component	Abbreviation	Function and Description
Generative Readiness Standard	GRS	The formal prerequisite checklist for initiating any cross-discipline generative activity. Requires confirmation of coherence (Layer 2),

Component	Abbreviation	Function and Description
		synchronization (Layer 3), and alignment (Layer 4) before a GAN may be issued.
Generative Initiative Register	GIR	A living register of all authorized, active, and completed generative initiatives. Maintained by the CDGL and reviewed each governance cycle. Records GAN issuance dates, initiative status, IAD reference, and GOC outcomes.
Integration Architecture Document	IAD	A per-initiative structural plan specifying the design of cross-discipline integration. Includes contributing disciplines, integration method, quality gates, submission format, and governance routing path. Produced by the CDGL in consultation with contributing disciplines.
Generative Output Standard	GOS	Quality criteria applied to all joint outputs prior to governance submission. Requires cross-discipline authorship, attribution, integration logic documentation, and demonstrated governance utility. Failure to meet GOS results in a Remediation Notice.
Generative Output Certificate	GOC	Formal attestation issued by the CDGL confirming that an output meets GOS. Required for governance consideration at Layer 6. A GOC is issued per output, not per initiative.
Emergence Register	ER	A structured log of unexpected generative outputs and emergent

Component	Abbreviation	Function and Description
		insights arising from authorized initiatives. Includes governance routing notes and, where applicable, Emergence Alert classifications. Reviewed by governance each cycle.

5. Operating Principles

The following five principles govern all CDGL operations. These principles are non-negotiable and inform the interpretation of every protocol, standard, and procedure within this document.

Principle 1 — Generative Integrity. Only outputs meeting the Generative Output Standard (GOS) may be submitted to governance as cross-discipline products. The CDGL bears institutional responsibility for the integrity of every output it certifies. GOS certification is not a procedural formality; it is a substantive determination.

Principle 2 — Attribution. The contribution of each discipline to every joint output is explicitly documented. No discipline's contribution is subsumed, merged without record, or erased in the integration process. Attribution records accompany all GOC-cleared outputs submitted to governance.

Principle 3 — Readiness Before Generation. No generative initiative is authorized before GRS prerequisites are confirmed. Initiating disciplines that proceed without a GAN operate outside Δ CS governance and their outputs are not eligible for GOC certification or governance submission.

Principle 4 — Emergence as Asset. Unexpected generative outputs are treated as institutional assets, not anomalies or deviations. The CDGL maintains systematic mechanisms for capturing, recording, and routing emergent outputs, and shall not suppress or delay their documentation.

Principle 5 — Governance Primacy. Generative outputs are in service of governance objectives. The CDGL does not generate outputs for its own

institutional purposes, expand its scope without governance authorization, or retain outputs that are governance-ready beyond the timelines specified in Section 9.

6. Generative Authorization Protocol

The following protocol governs all cross-discipline generative initiatives from proposal submission through governance delivery. All steps are mandatory. No step may be bypassed without explicit waiver authority from the Generative Director, which must be recorded in the GIR.

3. **Proposal Submission.** Initiating disciplines jointly submit a **Generative Initiative Proposal (GIP)** to the CDGL. The GIP shall specify: the generative objective, all contributing disciplines, the anticipated output type and format, the proposed timeline, and any known dependencies on lower-layer processes.
4. **GRS Review.** The CDGL reviews the GIP against the Generative Readiness Standard (GRS). Confirmation of readiness is formally requested from Layer 2 (Coherence), Layer 3 (Synchronization), and Layer 4 (Alignment). All three confirmations must be received before the initiative may proceed. Absent confirmation from any layer, the GIP is returned with a written statement specifying the unmet prerequisites.
5. **GAN Issuance and Registration.** Upon full GRS confirmation, the CDGL issues a **Generative Authorization Notice (GAN)** to all initiating disciplines and registers the initiative in the Generative Initiative Register (GIR), including the GAN issuance date and assigned Initiative Integration Lead.
6. **Integration Architecture Design.** The CDGL designs the **Integration Architecture Document (IAD)** in consultation with contributing disciplines. The IAD is finalized and signed by the Generative Director before generative work begins. Generative work may not commence prior to IAD finalization.

7. **Generative Execution.** The initiative proceeds under IAD governance. The assigned Initiative Integration Lead manages day-to-day cross-discipline generative work and reports progress to the CDGL at intervals specified in the IAD. The Emergence Curator monitors for emergent outputs throughout this phase.
8. **Output Submission and GOS Evaluation.** Upon completion, the contributing disciplines submit the finished output to the CDGL for GOS evaluation. The CDGL completes evaluation within 10 working days. Outputs satisfying all GOS criteria are issued a **Generative Output Certificate (GOC)**. Outputs failing GOS are returned with a Remediation Notice (see Section 11).
9. **Governance Submission.** The Governance Liaison (CDGL) submits all GOC-cleared outputs to the Governance-Level Evaluation Layer (Layer 6) for governance consideration. Submission is accompanied by the GOC, the IAD, and the attribution record. The GIR is updated to reflect completed status.

Protocol Integrity Note

The seven-step protocol above is sequential and non-recursive except where Remediation (Step 6) returns an output to contributing disciplines for revision and resubmission. A resubmitted output re-enters the protocol at Step 6 only. A second GOS failure triggers escalation under Section 11 rather than a second Remediation Notice.

7. Integration Methods

The Δ CS framework formally recognizes four integration methods. The Integration Architecture Document (IAD) for each initiative shall specify the applicable method. Only methods listed below are recognized for purposes of GOS certification. Hybrid or composite methods require Generative Director approval and must be documented in the IAD.

#	Method	Description	Appropriate Use
1	Convergent Integration	Each contributing discipline produces a parallel, independently developed output. The CDGL formally merges these parallel outputs into a unified deliverable, reconciling overlaps and resolving gaps.	Where disciplines operate on distinct but related problems whose outputs require a unified form for governance purposes.
2	Sequential Integration	Contributing disciplines engage in a defined sequence, with each discipline building directly on the output of the preceding discipline. Order of contribution is specified in the IAD and is binding.	Where disciplines have established dependency relationships and generative progress is inherently cumulative.
3	Co-generative Integration	All contributing disciplines work simultaneously within a shared generative space under active CDGL facilitation. Integration occurs in real time rather than through post-hoc merging.	Where the generative problem requires continuous mutual adjustment between disciplines and sequential or convergent methods would lose integration value.
4	Synthesis Integration	The CDGL receives discipline outputs and synthesizes them into a new structural form not proposed by any contributing discipline. The	Where the integration gap between disciplines is sufficiently large that no contributing discipline can provide the

#	Method	Description	Appropriate Use
		synthesis form is an original CDGL product, distinct from the discipline inputs.	bridging structure; where governance requires a unified framework rather than attributed contributions.

8. Interaction Rules

8.1 Inputs to the CDGL

The CDGL receives formally designated inputs from the following sources:

- **Cross-Discipline Alignment Layer (Layer 4)** — Alignment verification, confirming that contributing disciplines have achieved the alignment standard required for generative work.
- **Coherence Layer (Layer 2)** — Readiness confirmation, verifying that contributing disciplines meet coherence prerequisites for generative activity.
- **Synchronization Layer (Layer 3)** — Synchronization clearance, confirming that contributing disciplines are operationally synchronized for joint generative production.
- **Contributing Disciplines (direct)** — Generative Initiative Proposals (GIPs) and completed output submissions for GOS evaluation.

8.2 Outputs from the CDGL

The CDGL produces formally designated outputs routed to the following recipients:

- **Governance-Level Evaluation Layer (Layer 6)** — All GOC-cleared generative outputs, submitted with accompanying GOC, IAD, and attribution record. All Emergence Alerts for emergent outputs with direct governance implications.

- **Governance Deliberative Bodies** — Via Layer 6 as the exclusive governance routing channel; the CDGL does not submit directly to deliberative bodies without Layer 6 intermediation.
- **Emergence Register** — All emergent outputs arising from authorized initiatives, for governance review each cycle.
- **Contributing Disciplines** — Generative Authorization Notices (GANs), Remediation Notices upon GOS failure, and completed IADs.
- **Lower Layers (2, 3, 4)** — GIP return notices where GRS prerequisites are unmet, specifying the layer responsible for remediation.

8.3 Prohibited Interactions

The CDGL shall not:

- Submit outputs to governance bodies without a valid GOC;
- Bypass Layer 6 to deliver outputs directly to governance deliberative bodies;
- Initiate generative work on behalf of disciplines without a GIP and GRS confirmation; or
- Retain GOC-cleared outputs beyond the submission timelines specified in Section 9.

9. Cadence and Review

Activity	Cadence / Timeline
Generative Initiative Authorizations (GAN issuance)	Rolling — triggered by GIP submission; no fixed cycle. GRS review completed within 10 working days of GIP receipt.
GOS Evaluations	Completed within 10 working days of output submission. Extensions require Generative Director written approval, recorded in the GIR.
GOC-Cleared Output Submission to Layer 6	Within 5 working days of GOC issuance.
Emergence Register Review	Each governance cycle. Contents

Activity	Cadence / Timeline
	reported to Layer 6 for governance deliberation routing.
Generative Initiative Register (GIR) Review	Each governance cycle. Full status review of all active and pending initiatives.
CDGL Annual Performance Review	Annually. Submitted to the Governance-Level Evaluation Layer (Layer 6). Covers: initiatives authorized, outputs certified, GOS failures, emergence events, and remediation outcomes.
GRS and GOS Criteria Review	As part of the annual performance review cycle, or triggered by repeated GOS failures per Section 11.

10. Roles and Accountabilities

Role	Accountabilities
Generative Director (GD)	Leads the CDGL. Authorizes all generative initiatives (issues GANs). Certifies all generative outputs (issues GOCs). Designs or formally delegates the design of Integration Architecture Documents. Signs off on all IADs before generative work begins. Approves any waiver of protocol steps. Submits the CDGL Annual Performance Review to Layer 6.
Initiative Integration Leads (per initiative)	Appointed per authorized initiative. Manage day-to-day cross-discipline generative work under the governing IAD. Report progress to the CDGL at IAD-specified intervals. Coordinate contributing discipline participation. Flag emerging issues to the Generative Director.
Emergence Curator	Maintains the Emergence Register. Monitors all active generative processes for emergent outputs. Documents emergent outputs in the ER with governance routing notes. Issues Emergence Alerts to the Governance Liaison and Generative Director when

Role	Accountabilities
	emergent outputs carry direct governance implications.
Governance Liaison (CDGL)	Manages the formal submission of all GOC-cleared outputs to Layer 6. Coordinates with Layer 6 on governance routing. Maintains records of all Layer 6 submissions. Communicates Emergence Alerts to Layer 6 as directed by the Emergence Curator and Generative Director.

11. Escalation Pathways

The following escalation pathways apply to all deviations, failures, and emergent governance events arising within the CDGL. Escalation is mandatory where specified; discretionary escalation is at the Generative Director's determination.

Trigger Condition	Escalation Pathway
GIP rejected due to unmet GRS prerequisites	GIP returned to initiating disciplines with a written statement identifying the specific unmet prerequisites. Remediation request routed to the relevant lower layer(s) — Layer 2 (coherence deficiency), Layer 3 (synchronization deficiency), or Layer 4 (alignment deficiency). Resubmission of GIP may occur once prerequisites are confirmed.
GOS failure on completed output (first instance)	Output returned to contributing disciplines with a Remediation Notice specifying the GOS criteria unmet and the remediation required. Disciplines may revise and resubmit once. Resubmission re-enters the protocol at Step 6 only.
GOS failure on resubmitted output (second instance)	Escalated to the Governance-Level Evaluation Layer (Layer 6) for systemic review. The CDGL submits a written report detailing both GOS evaluations, the Remediation Notice issued, and the CDGL's assessment of the systemic cause. Layer 6 determines the resolution

Trigger Condition	Escalation Pathway
	pathway.
Emergent output with direct governance implications	Immediate routing to Layer 6 via an Emergence Alert , issued by the Governance Liaison on the Emergence Curator's direction. The Emergence Alert includes: a description of the emergent output, its assessed governance implications, and the CDGL's recommended routing. The Emergence Register is updated simultaneously.
Initiative proceeding without valid GAN	Immediate suspension of the initiative by the Generative Director. Initiating disciplines notified in writing. No output from the unauthorized initiative may be submitted for GOS certification or governance consideration. Incident recorded in the GIR.
GOS evaluation not completed within 10 working days	Generative Director must issue a written extension approval or initiate an expedited review. Unexplained delays reported to Layer 6 in the next cycle report.

12. Amendment Conditions

The instruments and protocols of the CDGL are subject to amendment under the following conditions. All amendments are recorded in the GIR with the effective date and authorizing authority.

Instrument	Amendment Authority	Conditions
Generative Readiness Standard (GRS) criteria	Governance supermajority, following CDGL review and recommendation	CDGL must submit a written review of the proposed amendment before any governance vote. Amendment may not reduce GRS prerequisites below Layer 2, 3, and 4 confirmation requirements.
Integration Methods (Section 7)	Addition: CDGL with governance confirmation.	New integration methods proposed by the CDGL

Instrument	Amendment Authority	Conditions
	Removal: governance supermajority only.	require governance confirmation before operational use. No method may be removed without a governance supermajority vote. Hybrid or composite methods require Generative Director approval per initiative (not a standing amendment).
Generative Output Standard (GOS) criteria	Governance supermajority, following CDGL review and recommendation	CDGL must submit a written review of the proposed amendment, including analysis of any GOS failure trends. Amendment may not reduce GOS below the four criteria specified in Section 3.3.
Integration Architecture Document (IAD) format	CDGL Director, with Generative Director sign-off	Format amendments apply prospectively to new initiatives only. Active initiatives retain the IAD format in effect at GAN issuance unless amended by mutual agreement of contributing disciplines and Generative Director.
Generative Authorization Protocol (Section 6)	Governance supermajority, following CDGL review	No step may be removed from the seven-step protocol without governance supermajority approval. New steps may be added by governance simple majority on CDGL recommendation.
This document (Δ CS-GOV-L5-CDGL) in its entirety	Governance supermajority	Full document revision requires governance supermajority. Revised versions supersede all prior versions; prior version archived with effective dates recorded.

Amendment Record Requirement

All amendments to any CDGL instrument must be recorded in the Generative Initiative Register (GIR) with the following information: instrument amended, prior version or criteria, amended version or criteria, authorizing body, authorization date, and effective date. The Generative Director is responsible for maintaining the currency of this document and its referenced instruments.

Document ID: Δ CS-GOV-L5-CDGL | **Version:** 1.0 | **Status:** Active | **Effective Date:** July 2026

Revision Tracking: All revisions to this document must increment the version number (e.g., 1.0 $\rightarrow$ 1.1 for minor amendments; 1.x $\rightarrow$ 2.0 for full document revision), update the effective date, and record the authorizing body and date of authorization. A complete revision history shall be maintained as an appendix to the current version. Superseded versions are archived and remain accessible to governance bodies upon request. The Generative Director holds responsibility for version currency and distribution control.

Distribution: Restricted to Δ CS governance participants, CDGL role-holders, and authorized Layer 6 reviewers. Redistribution requires Generative Director authorization.

GOVERNANCE FRAMEWORK — RESTRICTED | Δ CS-GOV-L5-CDGL v1.0 | © Δ CS Framework Governance Body